

MSTAR PRACTICE WORKSHEET**Lesson 3-3: Determine Equivalent Ratios Using Tables**

Grade 6 Mathematics · Standard 6.AT.3a

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

A recipe uses 3 eggs for every 5 cups of flour. How many eggs are needed for 15 cups of flour?

- ☐ A 6
- ☒ B 9
- ☐ C 10
- ☐ D 12

The scale factor is $15 \div 5 = 3$. Multiply the eggs by 3: $3 \times 3 = 9$ eggs.

Question 2 **SELECTED RESPONSE — CORRECT: A**

Which row does NOT belong in a ratio table for 4:5?

- ☒ A 6:10
- ☐ B 8:10
- ☐ C 12:15
- ☐ D 16:20

6:10 simplifies to 3:5, not 4:5. The other ratios (8:10, 12:15, 16:20) all simplify to 4:5.

Question 3 SELECTED RESPONSE — CORRECT: B

A ratio table shows 3:7, 6:14, and 9:21. What is the next row?

- ☐ A 10:24
- ☒ B 12:28
- ☐ C 12:24
- ☐ D 15:28

The pattern adds 3 to the first value and 7 to the second each time. After 9:21: $9+3=12$ and $21+7=28$. The next row is 12:28.

Question 4 SELECTED RESPONSE — CORRECT: B

A ratio table for 2:5 starts with 2:5, 4:10. What is the third row?

- ☐ A 5:10
- ☒ B 6:15
- ☐ C 6:12
- ☐ D 8:15

The additive pattern adds 2 to the first value and 5 to the second. $4+2=6$ and $10+5=15$. The third row is 6:15.

Question 5 SELECTED RESPONSE — CORRECT: B

In a ratio table, the ratio of sugar to butter is 3:2. If you use 12 cups of sugar, how many cups of butter do you need?

- ☐ A 6
- ☒ B 8
- ☐ C 9
- ☐ D 10

Scale factor: $12 \div 3 = 4$. Multiply butter by 4: $2 \times 4 = 8$ cups of butter.

Question 6

SELECTED RESPONSE — CORRECT: B

A ratio table shows 5:8, 10:16, 15:24. What scale factor was used from the first row to the third row?

- ☐ A 2
- ☒ B 3
- ☐ C 5
- ☐ D 8

From 5 to 15: $15 \div 5 = 3$. From 8 to 24: $24 \div 8 = 3$. The scale factor is 3.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: B

A ratio table pairs 3 cups of flour with 2 cups of water. If the recipe is scaled to use 12 cups of flour, how much water is needed?

- ☐ A 6 cups
- ☒ B 8 cups
- ☐ C 11 cups
- ☐ D 24 cups

$12 \div 3 = 4$, so the scale factor is 4. Both terms scale: $2 \times 4 = 8$ cups of water.

- **A:** That adds 4 to the water instead of scaling it. Flour was MULTIPLIED by 4, so water is too.
- **C:** That adds the same 9 that was added to the flour. Ratio tables scale by multiplying, not adding.
- **D:** That scales the flour term. The scale factor is 12 divided by 3, applied to the 2 cups of water.

Part B — correct answer: C

Why is 11 cups an incorrect answer for Part A?

- ☐ A It divides instead of multiplying.
- ☐ B It uses the wrong conversion factor.
- ☒ C It adds 9 to both terms instead of multiplying both by the same factor.
- ☐ D It reverses flour and water.

Adding the same amount to both terms changes the relationship. Equivalent ratios require multiplying or dividing BOTH terms by the same non-zero factor.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about ratio tables and equivalent ratios:

- ☒ **A** Multiplying both terms of a ratio by 3 produces an equivalent ratio.
- ☒ **B** Dividing both terms by a common factor produces an equivalent ratio.
- ☐ **C** Adding 5 to both terms produces an equivalent ratio.
- ☒ **D** Every row of a correct ratio table simplifies to the same ratio.
- ☒ **E** A ratio table can be extended in either direction, up or down.

A, B, D, and E are true. C is false — equivalence comes from multiplying or dividing both terms, never from adding.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A recipe uses 4 cups of flour for every 3 cups of sugar. A student filled in a ratio table: 4:3, 8:6, 12:9, 14:12. Find and fix the error in the last row. Explain what went wrong.

Model answer: The last row is wrong. The pattern multiplies by the same scale factor. $4 \times 1 = 4$, $4 \times 2 = 8$, $4 \times 3 = 12$, so the 4th row should be $4 \times 4 = 16$ cups of flour and $3 \times 4 = 12$ cups of sugar $\rightarrow 16:12$. The student added 2 to 12 instead of adding 4. Or: $12 + 4 = 16$, not $12 + 2 = 14$.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.